

A Maki-Thompson model with updated and verified information

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ALTENCOA11-2026, 5 al 9 de octubre, 2026

San Juan de Pasto, Universidad de Nariño

Abstract. Building on classical frameworks such as the Daley–Kendall (DK) and Maki–Thompson (MT) models, this lecture introduces a novel generalization of rumor propagation. We propose a model that incorporates two key innovations: in one hand, firstly, it extends the MT model by allowing multiple types of spreaders and permits stiflers to become spreaders again without returning to an ignorant state and secondly, it models updating, distortion and verification of the information. When a stifier is reactivated as a spreader, they randomly flip one bit of the received information. Additionally, spreaders can verify information at any time and, upon confirmation of inaccuracy, correct the transmitted information to match the true version. This framework captures the complexity of real-world rumor evolution, including partial information, recirculation of rumors, and the mitigating role of fact-checking.

Keywords. Maki-Thompson model, updating, information, distortion, verification.

Information exchange and rumor sharing are essential aspects of human social interaction. The dissemination of information and rumors can have significant effects on society, as well as on its political and economic structures. Information can influence public opinion on political conjunctures [2], impact financial markets [4,5], create social tensions and unnecessary generalized anxiety about possibly false public issues [7]. Models that study and help to understand the dissemination of rumors could be crucial to combat the damage caused by misinformation and, above all, fake news. Within this framework, the ability to verify information emerges as an essential instrument.

The standard model for rumors was introduced by Daley and Kendall in the mid-1960s [1]. They proposed a simple mathematical model for rumor transmission, which emerged as an alternative to the well-known susceptible-infected-removed (SIR) epidemic model. The population considered is finite, homogeneously connected, and subdivided into three classes of individuals: *ignorant* (people who do not know the rumor), *spreaders* (people who know the rumor and actively try to spread it) and *stiflers* (people who know the rumor but no longer spread it). Once a rumor is introduced into the population, it spreads according to specific rules that depend on the class of the individual who knows the rumor. In the Daley–Kendall (DK) model, the rumor propagates through pairwise contacts between spreaders and other individuals. A spreader attempts to transmit the rumor to others. During a contact between a spreader and an ignorant individual, the ignorant becomes a spreader; in all other types of contact, the individuals involved become stiflers. This model was later modified by Maki and Thompson in 1973 [6], who changed the rules of rumor transmission. In the Maki–Thompson (MT) model, only the initiating spreader becomes a stifier.

In this lecture, we will define our model and show some results. In the model, we introduce two main innovations: first, it generalizes the MT model by incorporating multiple types of spreaders and allowing stiflers to act as spreaders again; second, it takes into account the potential distortion

and verification of the transmitted information.

Consider a rumor composed of $m \in \mathbb{N}$ concatenated bits (zeros and ones). If an individual is a spreader, they can transmit any of the $|\mathcal{P}(m)| - 1$ distinct parts of the rumor, where $\mathcal{P}(m)$ denotes the power set of $1, 2, \dots, m$. The empty set $\emptyset$ is excluded, as it would represent no information to transmit. This new mechanism generates different types of spreaders and it is used to update information among spreaders. Our model also considers the possibility for stifflers to spread the rumor again. A stiffler can become a spreader if they had been persuaded by a spreader to transmit information once more. It is at this point that the possibility of information distortion is introduced. When a stiffler is convinced to transmit information again, they randomly select one symbol from the received information and flip it. Verification of the information could be done at any moment during the dynamic process. With a given rate, spreaders can verify that the information they are spreading is accurate, and then decide whether to alter the transmitted information to match the true information. This behavior intuitively occurs when a person decides not to completely trust the information they have and seeks a trustworthy source.

This is joint work with *Silvio da Costa Ferreira Junior and Thainá Ferreira from Universidade Federal de Viçosa, Brazil and Yamir Moreno from Universidad de Zaragoza in Spain.*

Acknowledgements

This study was supported by grant 2024/15437-0 São Paulo Research Foundation (FAPESP).

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